

UE25CS251A
DIGITAL LOGIC AND COMPUTER DESIGN

TEAM DLCD

Department of CSE and CSE (AI & ML)

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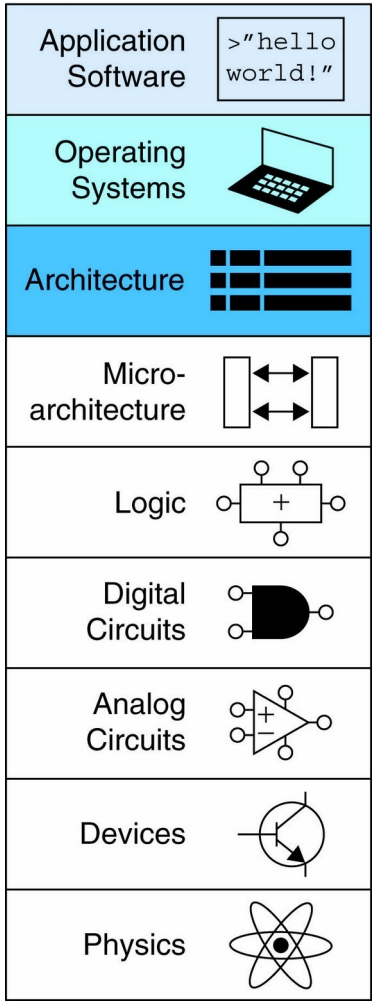
Sequential Building Blocks

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TOPICS

- **Counters**
- **Shift Registers**



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Sequential Building Blocks

Sequential building blocks are digital circuits that **store information and change their state in response to a clock**.

Unlike combinational circuits, whose outputs depend only on their current inputs, sequential circuits have **memory**. This memory is generally implemented using flip-flops.

Two important sequential building blocks discussed in this section are:

1. Counters

A counter stores a binary value and changes that value according to a fixed counting sequence. A binary counter normally increments its stored value by 1 on every rising edge of the clock.

2. Shift Registers

A shift register stores multiple bits and moves the stored bits from one flip-flop to another on every clock edge.

Applications

- Counters → timing, event counting, frequency division
- Shift registers → serial/parallel data conversion and digital testing

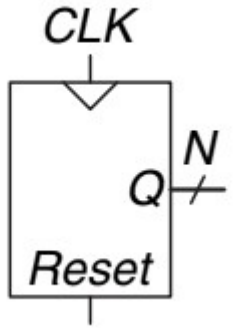
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N-bit Binary Counter

What is a Counter?

An **N-bit binary counter** is a sequential arithmetic circuit with:



- A **clock input**
- A **reset input**
- An **N-bit output Q**

When reset is asserted, the counter is initialized to zero. After reset is released, the counter **increments by 1 at every rising edge of the clock**.

General operation: $Q_{next}=Q+1$

Therefore, if the current value is: Q=0101, then at the next rising clock edge: Q=0110

The counter continues this process until it reaches its maximum value and then wraps around to zero.

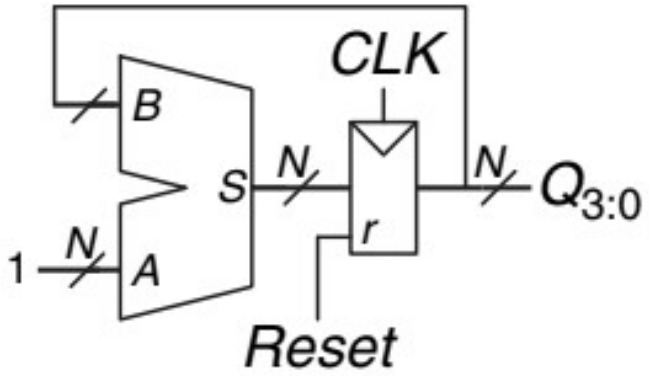
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N-bit Binary Counter

Figure here, shows an **N-bit counter composed of an adder and a resettable register**.

On each cycle, the counter adds 1 to the value stored in the register.



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Number of States in an N-bit Counter

How many values can an N-bit counter represent?

An N-bit binary number can represent: **2^N** different combinations.

Therefore, an N-bit counter has: 2^N possible states

The values range from: 0 to 2^(N−1)

Example: 3-bit counter

Since: 2^3=8

the counter has 8 states:

000 → 001 → 010 → 011
↑ ↓
111 ← 110 ← 101 ← 100

After 111, the next value is: 000 because the counter has exhausted all 8 possible 3-bit combinations.

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Number of States in an N-bit Counter

Example: 8-bit counter

2^8=256

So an 8-bit counter counts from: 0 to 255, before returning to 0.

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Reset and Clock

The N-bit binary counter has:

- **Clock input**
- **Reset input**
- **N-bit output Q**

Reset

Reset initializes the output to 0.

Clock

The counter increments on the:

rising edge/falling edge of the clock

Thus: Reset→Q=0 and subsequently, on each rising edge: Q→Q+1

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Types of counters

- **Asynchronous Counter(Ripple Counter)**
- **Synchronous Counter**

Counter Type	Clocking Method	States	Speed	Applications
Asynchronous	Ripple (chained)	2^N	Slower	Simple event counting
Synchronous	Global clock	2^N	Faster	High-speed systems
Up	Increment only	2^N	Normal	Timers, counters
Down	Decrement only	2^N	Normal	Countdown timers
Up/Down	Bidirectional	2^N	Flexible	Digital clocks
Decade (MOD-10)	Reset at 10	10	Normal	Frequency division
MOD-N	Reset at N	N	Normal	Custom sequences
Ring	Circulating '1'	N	Fast	Sequence control
Johnson	Inverted feedback	2N	Fast	Pattern generation

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Types of counters-Differences

Feature	Synchronous Counter	Asynchronous Counter
Clocking	All flip-flops receive the clock signal simultaneously	Flip-flops receive clock from the previous stage (ripple effect)
Speed	Faster due to simultaneous switching	Slower due to propagation delay
Propagation Delay	Minimal	Increases with more flip-flops
Complexity	More complex circuit with extra logic gates	Simpler design, fewer components
Power Consumption	Higher due to additional gates	Lower due to fewer components
Reliability	More reliable and stable	Prone to glitches and timing issues
Cost	Higher due to extra hardware	Lower as it requires fewer components
Applications	Used in high-speed operations	Suitable for low-speed applications

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Types of counters-Differences

Why T and JK Flip-Flops Are Preferred

- **T Flip-Flop (Toggle Flip-Flop)**
  - Has a single input (T).
  - When T = 1, the output toggles on each clock edge.
  - This directly divides the clock frequency by 2, making it ideal for counters and frequency dividers.
  - Simple design, minimal logic required.
- **JK Flip-Flop**
  - More versatile: can perform **Set, Reset, Hold, and Toggle** depending on J and K inputs.
  - When J = K = 1, it behaves exactly like a T flip-flop (toggle mode).
  - Eliminates the invalid state problem of SR flip-flops.
  - Widely used in counters because of flexibility and reliability.

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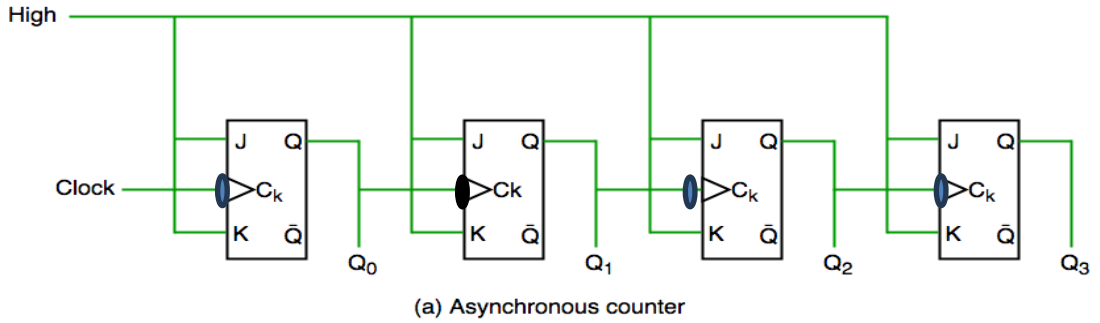
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Types of counters-Asynchronous Counter

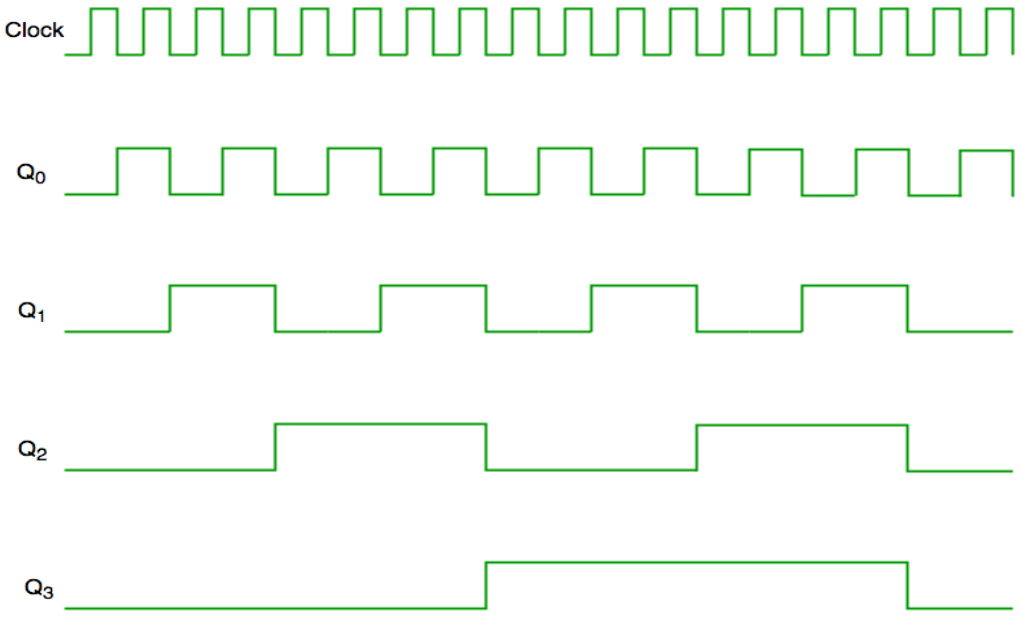
Asynchronous Counter (4 BIT RIPPLE / MOD 4 COUNTER)

In an asynchronous counter, there is no common (universal) clock for all flip-flops; only the first flip-flop is driven by the main clock, and each subsequent flip-flop is triggered by the output of the previous one, causing the count to ripple through the circuit.

From the timing diagram,
Q0 changes immediately at the falling edge of the clock,
Q1 changes at the falling edge of Q0, and the same pattern continues for Q2 and Q3.
This step-by-step change creates a ripple effect through the flip-flops, **which is why it is called a ripple counter or serial counter**. A ripple counter is a cascade of flip-flops where the output of one flip-flop acts as the clock input for the next.



(a) Asynchronous counter



(b) Timing Diagram

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Asynchronous Counter-MOD-4

Clock Pulse	Q_3	Q_2	Q_1	Q_0	Decimal Value
Reset	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
…	…	…	…	…	…
15	1	1	1	1	15
16	0	0	0	0	0 (wraps)

Timing Diagram (as in the image)

- **Clock**: square wave pulses.
- **Q_0**: toggles every pulse.
- **Q_1**: toggles every 2 pulses.
- **Q_2**: toggles every 4 pulses.
- **Q_3**: toggles every 8 pulses.
- Together, they form a binary count sequence from 0 → 15.

Key Observations:

- This is a **MOD-16 counter** (because $2^4 = 16$states).
- It's **asynchronous**, so changes ripple through flip-flops with slight delays.
- Useful for **frequency division**:
  - Q_0 = clock ÷ 2
  - Q_1 = clock ÷ 4
  - Q_2 = clock ÷ 8
  - Q_3 = clock ÷ 16

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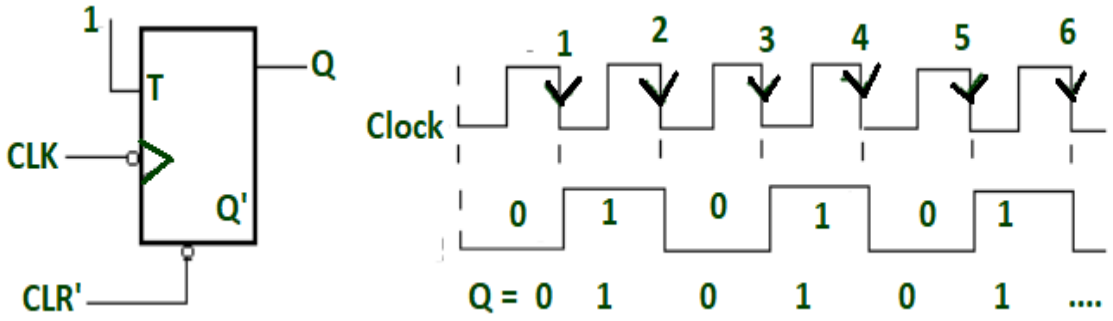
Asynchronous Counters -Examples

MOD-2 COUNTER(1-BIT)

When -ve edge clock pulse is applied and input is given to FF logic 1 then the output state of FF will toggle for every falling edge. The output frequency will be f/2 (If f is clock frequency).

It is known as binary or mod -2 counter or bit ripple counter.

It has 2 unique output states (0 and 1).

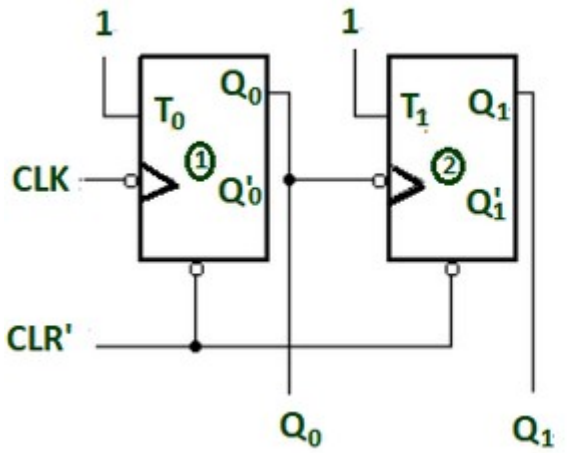


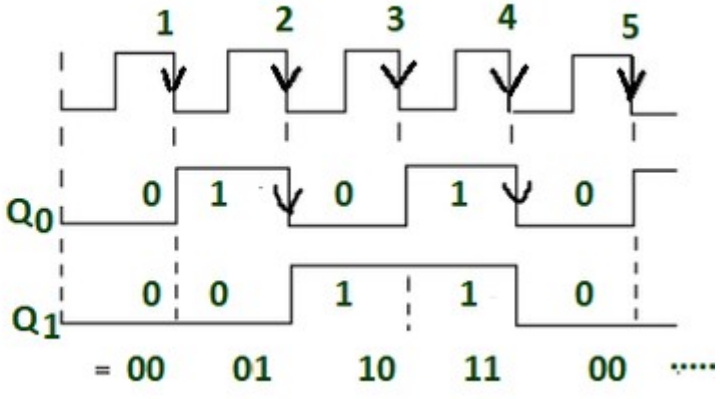
MOD 4 COUNTER(2-BIT)

When two FFs are connected in series and output of one FF is act as clock for 2nd FF. So the state of 2nd FF will change only when output and 1st FF is logic 1 and falling edge occur. The output frequency of Q_1 is f/4(if f is clock frequency).

It can generate 4 different unique states. This is known as divide by 4 circuits or mod 4 ripple counter.

Here output is taken as Q_1(MSB) Q_0(LSB).



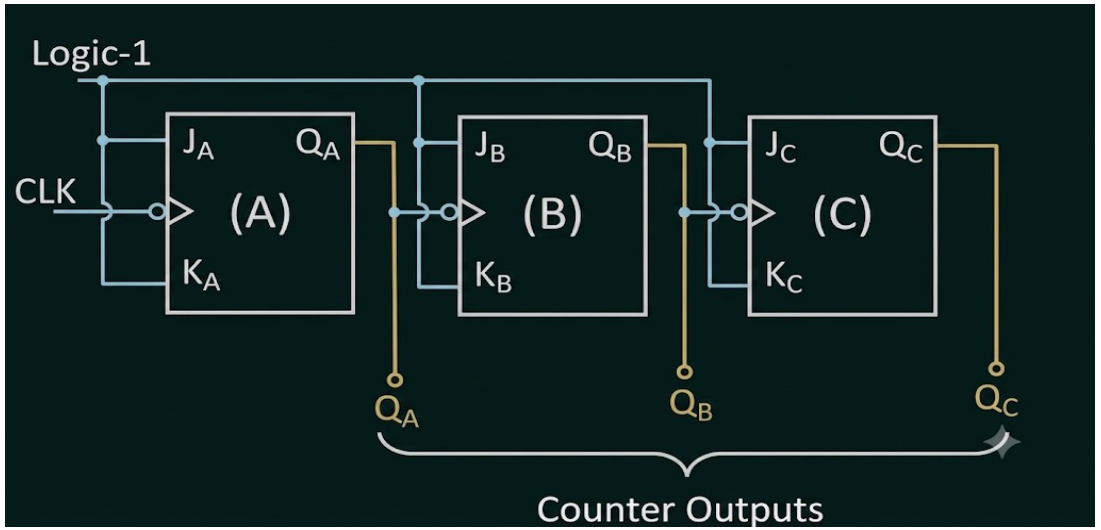


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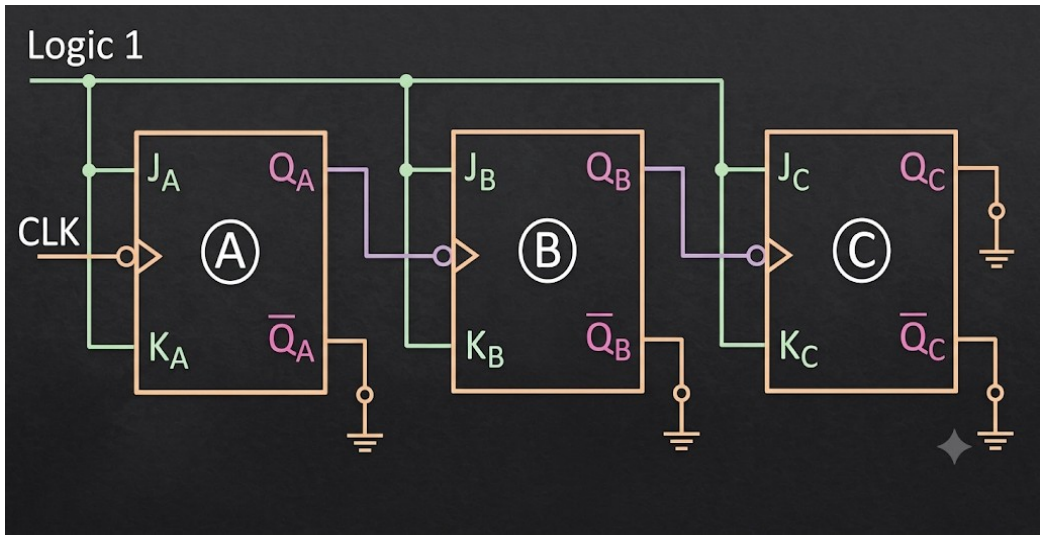
Asynchronous Counters - Examples

3 Bit-UP COUNTER



Counter State	Q_C	Q_B	Q_A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

3 Bit-DOWN COUNTER



Counter State	Q'_C	Q'_B	Q'_A
7	1	1	1
6	1	1	0
5	1	0	1
4	1	0	0
3	0	1	1
2	0	1	0
1	0	0	1
0	0	0	0

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Asynchronous Counters -Examples

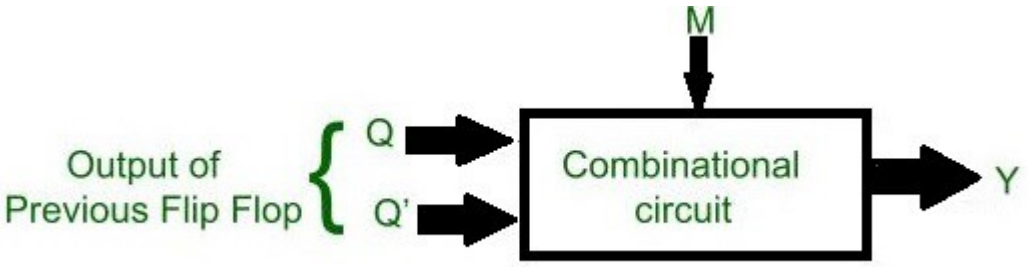
Design of 3 Bit Asynchronous UP/DOWN Counter
It is used more than separate up or down counter.

1. In this a mode control input (say M) is used for selecting up and down mode.

2. A combinational circuit is required between each pair of flip-flop to decide whether to do up or do down counting.

Step 1 : **Decision for Mode control input -**

Decision for mode control input
When M = 0, then Y= Q, therefore it will perform Up counting (As discussed above).
When M = 1, then Y= Q' therefore it will perform Down counting (As discussed above).
Combinational circuit is required for deciding mode control(i.e., whether counter will perform Up counting or Down counting).
So the all possible combinations are -



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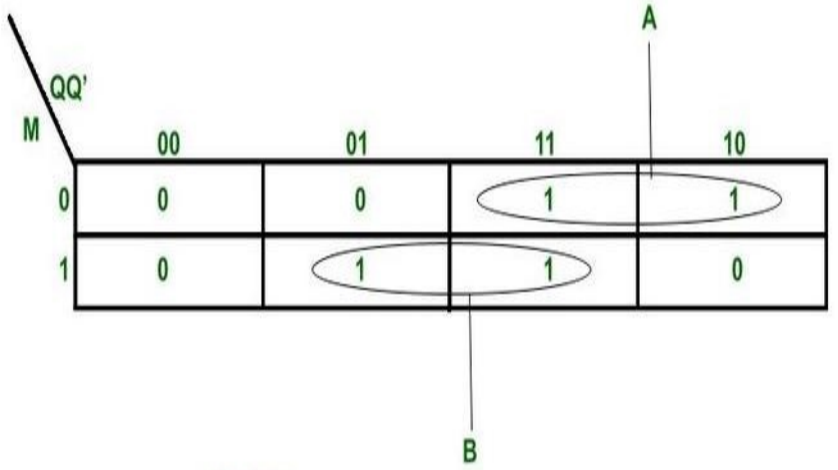
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Asynchronous Counters -Examples

Y = Q when M= 0
Y = Q' when M=1

M	Q	Q'	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Here Y is the output function



Y = A+B

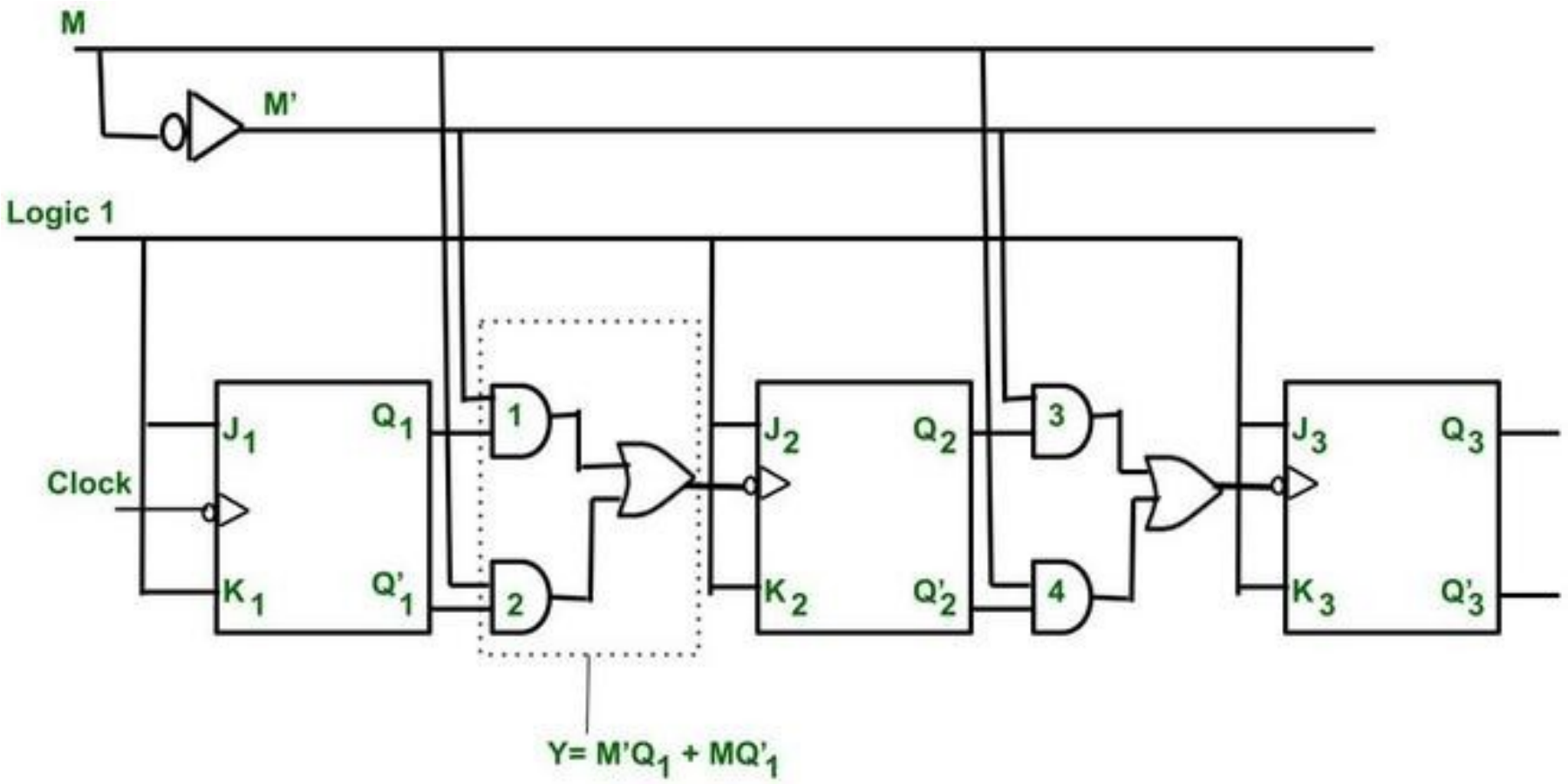
Y= M'Q + MQ'

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Asynchronous Counters -Examples

Insertion of Combinational logic between every pair of FFs -



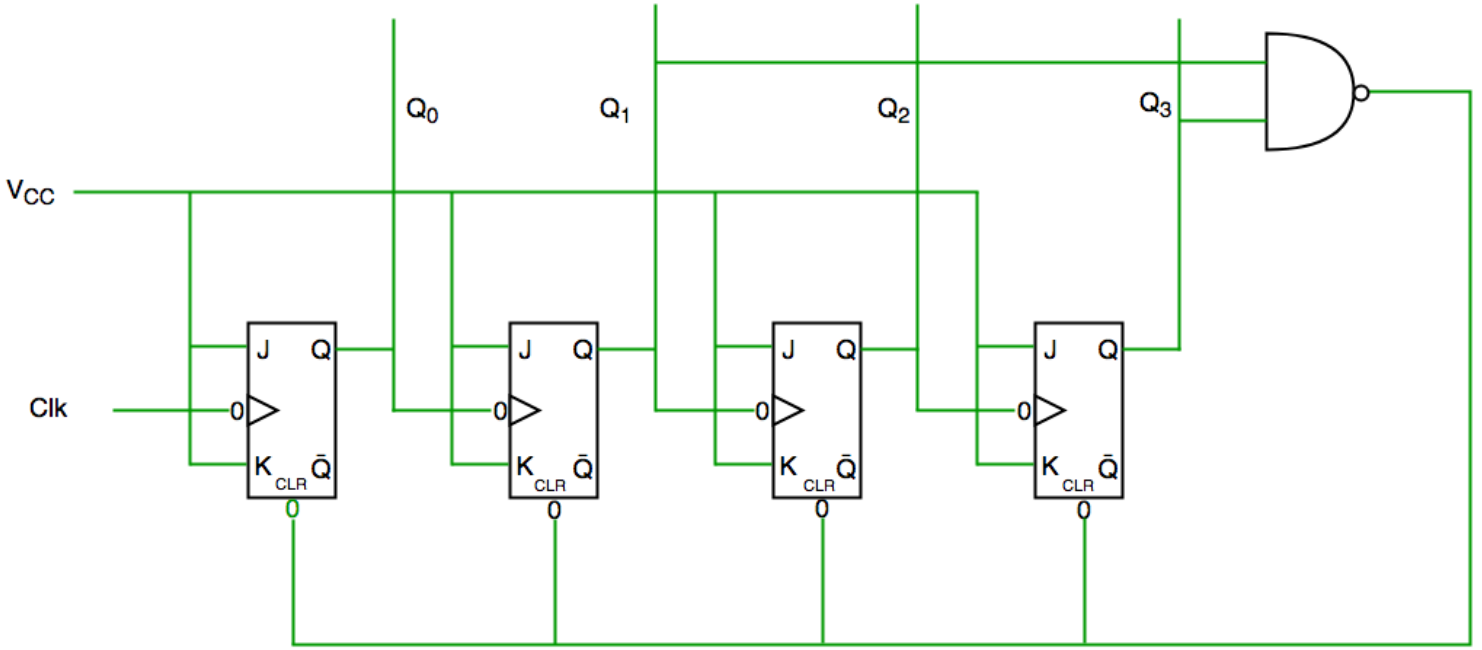
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Asynchronous Counters -Examples

Decade Counter

Counts ten states and then resets to its initial state. The most common decade counter counts from 0 to 9, but it can also be designed to count any ten states within 0 to 15 in a 4-bit counter before resetting.



Clock pulse	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

From the circuit diagram, a NAND gate is connected to Q3 and Q1 and its output is fed to the clear input because the binary representation of 10 is 1010. When Q3 and Q1 both become 1, the NAND gate output goes low, activating the clear input, so the counter resets at 10 and starts counting again from 0.

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Synchronous Counters

Synchronous Counter

Unlike an asynchronous counter, a synchronous counter uses a single global clock that drives all flip-flops simultaneously, so all outputs change in parallel. Its main advantage is that it can operate at higher frequencies because it does not suffer from cumulative delay, as the same clock is applied to every flip-flop. Hence, **it is also called a parallel counter.**

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Synchronous Counters -Examples

Steps to design Synchronous Counters

Step 1: Decide the number of flip flops.

Step 2: Excitation table of FF.

Step 3: State diagram and circuit excitation table.

Step 4: Obtain simplified equations using K' map.

Step 5: Draw the logic diagram.

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Synchronous Counters -Examples

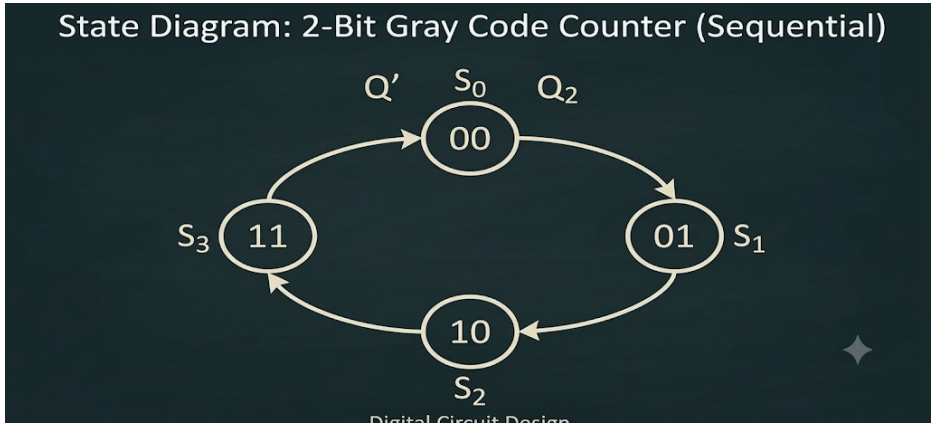
Design 2 bit Synchronous Up Counter

Step1: Decide the number of Flipflops – 2 (JK)

Step2: Excitation table of JK FF

INPUT		OUTPUT	
Qn	Q(n+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step3: State Diagram & Circuit ET



CKT Excitation Table : -

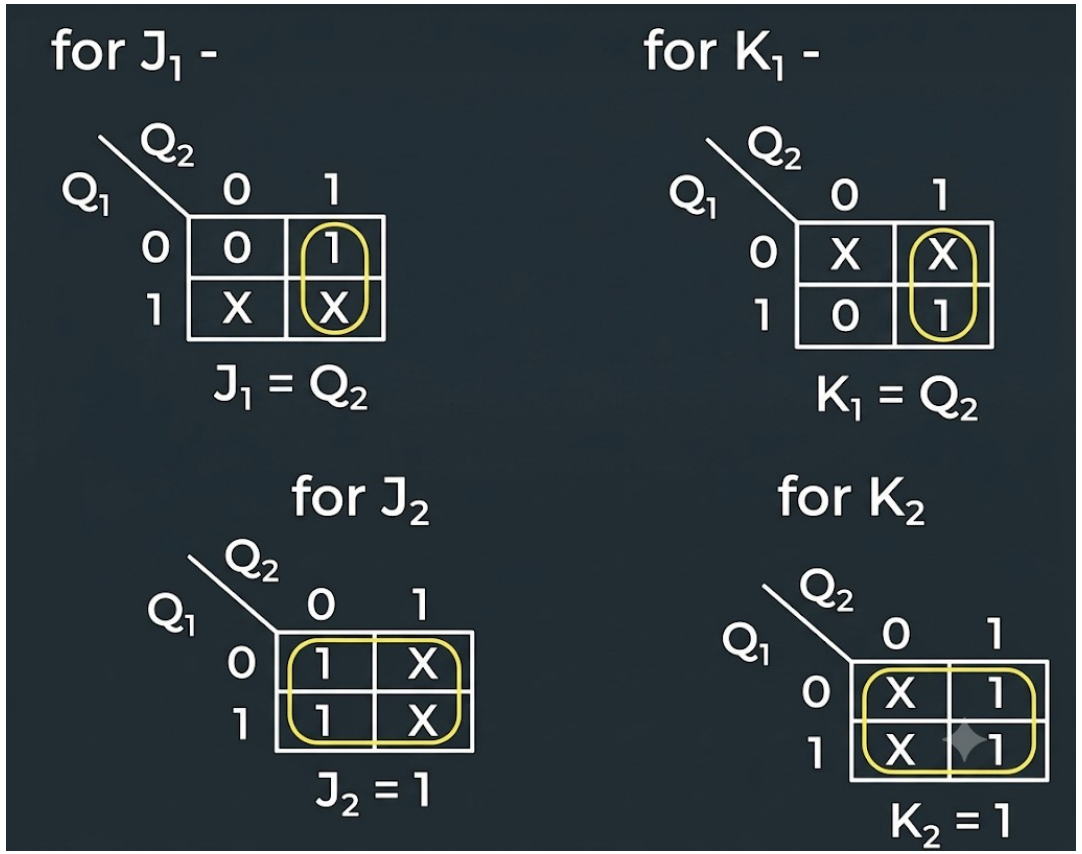
Q_1	Q_2	Q_1^*	Q_2^*	J_1	K_1	J_2	K_2
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

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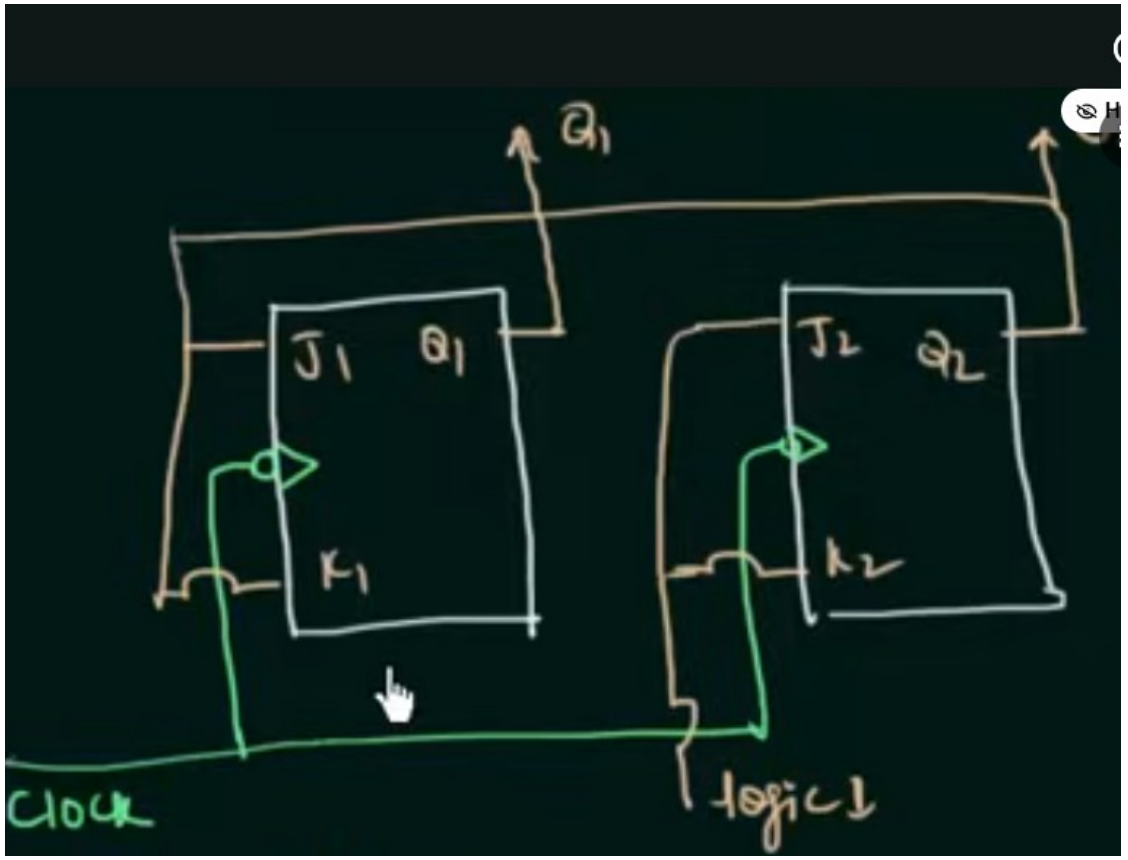
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Synchronous Counters -Examples

Step 4: Solve for J1 K1 J2 K2 USING K MAPS



Step 5: Draw the logic diagram



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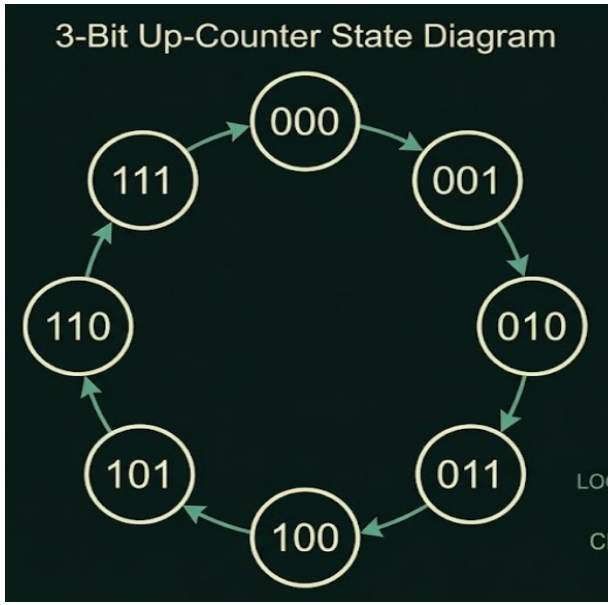
Synchronous Counters -Examples

Design 3 bit Synchronous Up Counter using T FF

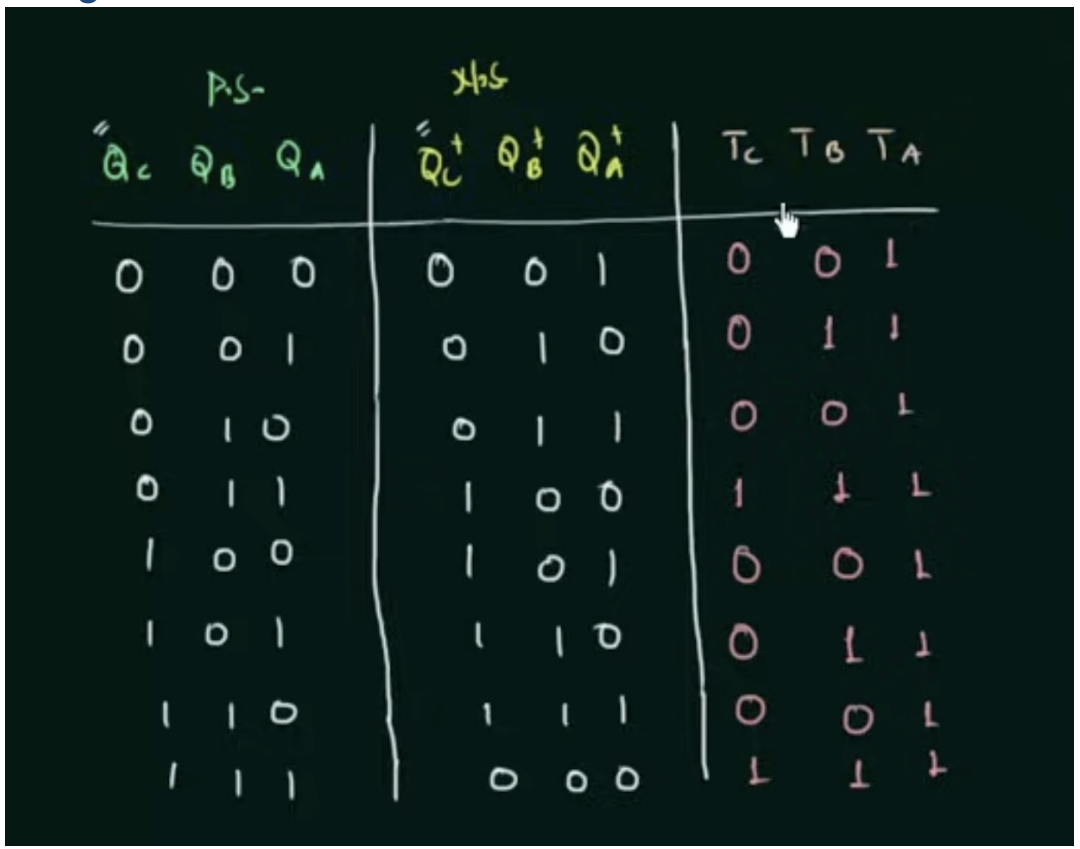
Step1: Decide the number of Flipflops – 3 (T FF)

Step2: Excitation table of T FF

INPUT		OUTPUT
Qn	Qn+1	T
0	0	0
0	1	1
1	0	1
1	1	0



Step3: State Diagram & Circuit ET

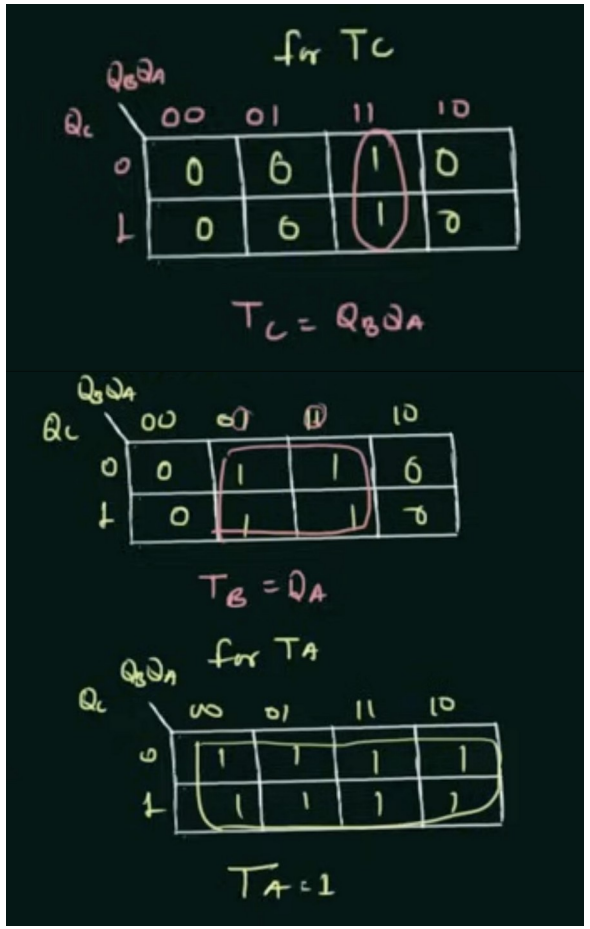


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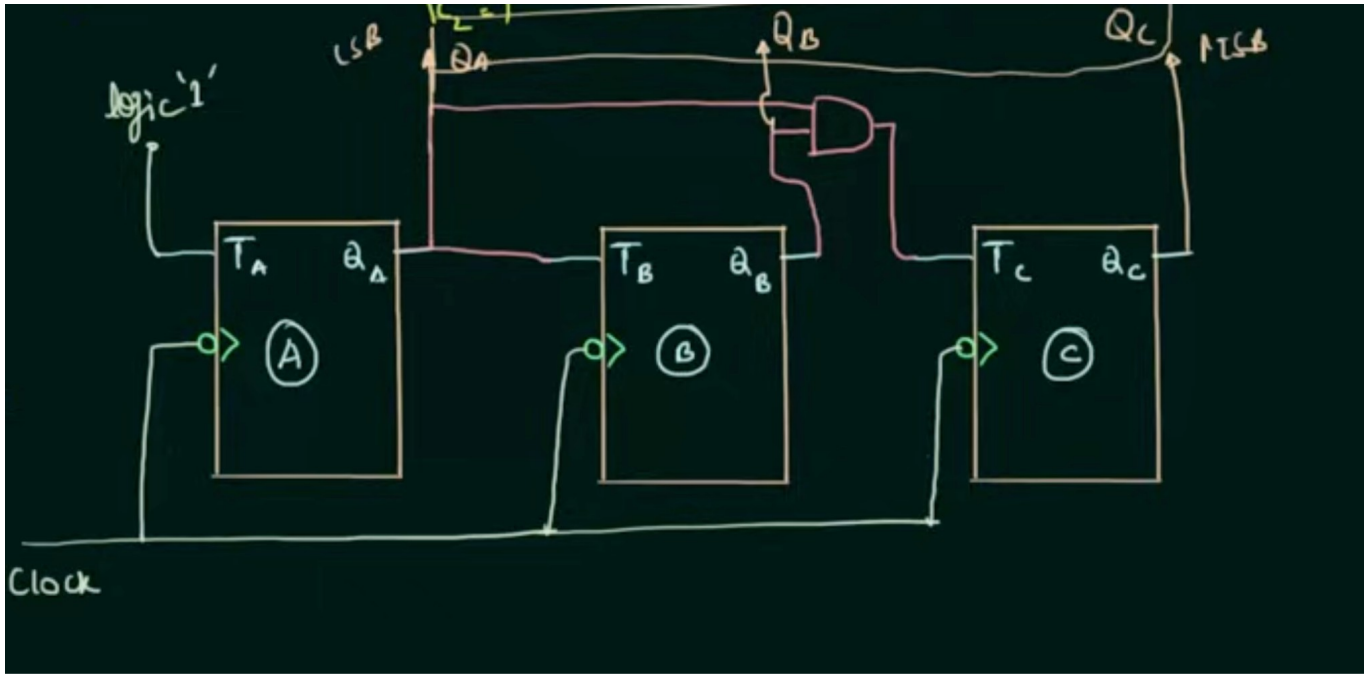
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Synchronous Counters -Examples

Step 4: Solve for T1 T2 T3 USING K MAPS



Step 5: Draw the logic diagram

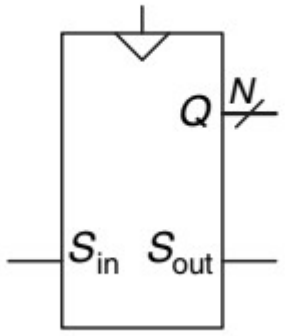


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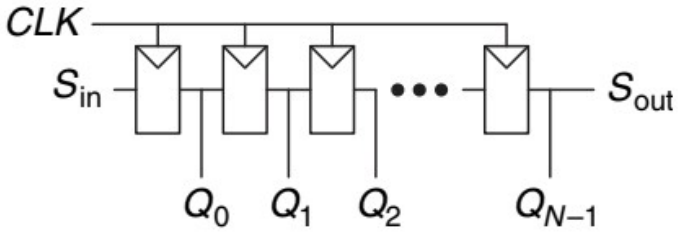
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What Is a Shift Register?

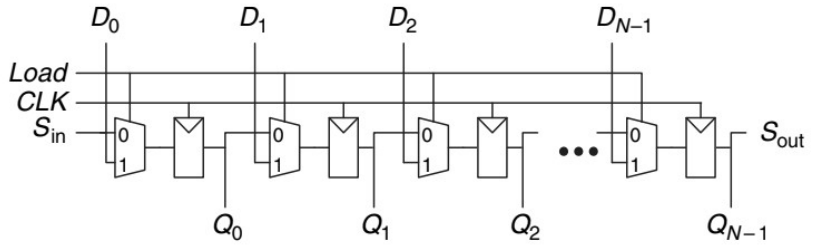
- A chain of flip-flops sharing one clock.
- On each clock edge, a new bit enters at Sin; every stored bit moves forward one position.
- The oldest bit exits at Sout.
- After N cycles, all N bits sit together on parallel outputs Q0…QN-1.
- **Function: converts a serial (one-bit-at-a-time) stream into a parallel N-bit word.**



Shift register symbol



Shift register schematic

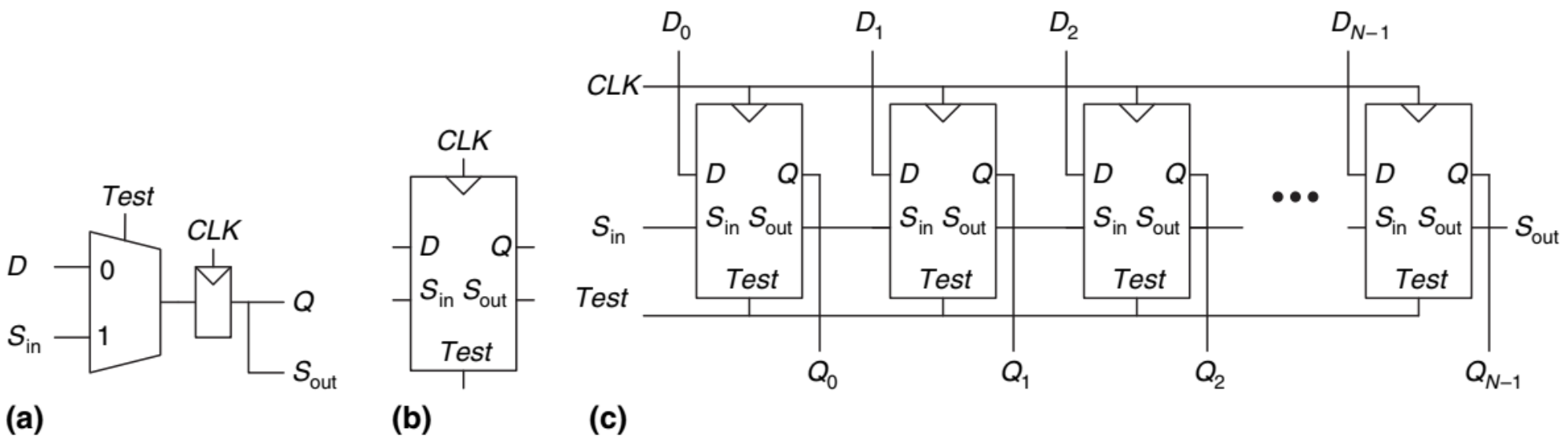


Shift register with parallel load

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Shift Registers



Scannable flip-flop: (a) schematic, (b) symbol, and (c) N-bit scannable register

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Shift Register

Internal structure:

- Built from N flip-flops connected in series.
- Each flip-flop's output feeds the next flip-flop's input.
- All flip-flops share the same clock signal.
- Some designs add a reset signal to clear all flip-flops to 0.

Adding Parallel Load:

- Basic shift register only does serial-to-parallel.
- Add a parallel data input D(N-1:0) and a control signal Load to also support parallel-to-serial.
- When Load = 1: all flip-flops load directly from D (parallel load, one cycle).
- When Load = 0: register shifts normally (Sin feeds in, bits move forward).

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HDL Description

```
always @(posedge clk or posedge rst)
  if (rst)       q <= 0;
  else if (load)  q <= data_in;        // parallel load
  else           q <= {q[N-2:0], sin}; // shift in sin

assign sout = q[N-1];
```

- rst clears everything, highest priority.
- load loads the full word from data_in in one cycle.
- Otherwise, bits shift up and sin fills the empty slot.
- sout is always the top bit of the register.

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Why Test Sequential Circuits Differently ?

- Combinational circuits: easy to test, apply input, check output.
- Sequential circuits: hard since the output depends on internal state.
- Example: to verify bit 31 of a 32-bit counter flips 0→1, naively need to reset then apply ~2 billion clock pulses.
- Problem: too many flip-flops to give each one its own test pin.

Scan Chains: The Solution

- Connect all flip-flops in the system into one long shift register called a scan chain.
- Normal mode: each flip-flop loads from its regular D input; scan connections ignored.
- Test mode: flip-flops shift data serially in/out via Sin/Sout — internal state can be read out or overwritten directly.
- The mux for switching modes is usually built into the flip-flop itself, forming a "scannable flip-flop."

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Type of counters - Ring counters

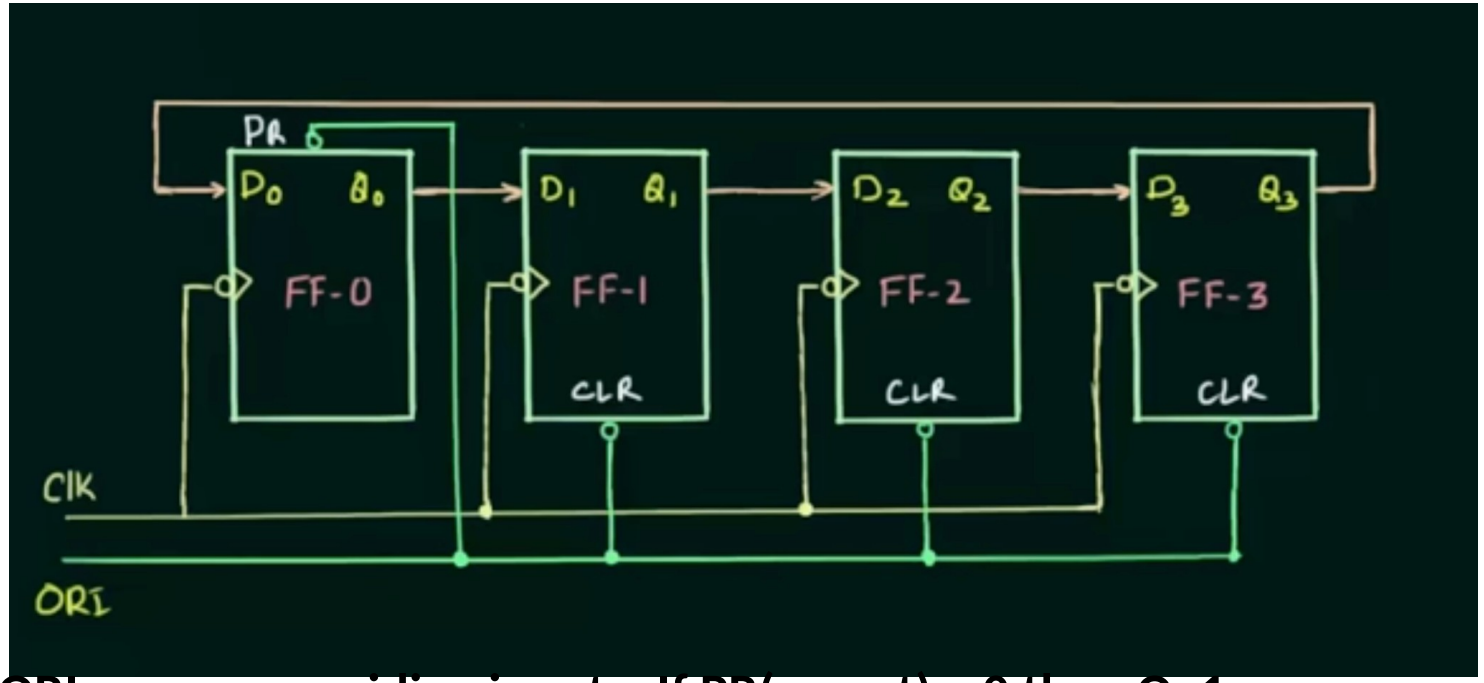
Definition

- A ring counter is a **shift-register-based counter** in which the output of the last flip-flop is fed back to the first.
- It consists of **n flip-flops connected in a circular manner**.
- A single 1 circulates through the flip-flops with each clock pulse.

Example: 4-bit Ring Counter

1000 → 0100 → 0010 → 0001 → 1000 → …

Key Point:
For **n flip-flops → n valid states → MOD-n counter**



ORI means overriding input: If PR(preset) = 0 then Q=1

If Clr = 0 then Q=0

It is used to initialize counters as 1 0 0 0

It is used to **initialize (preset) the ring counter to a valid starting state**.

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Type of counters - Ring counters

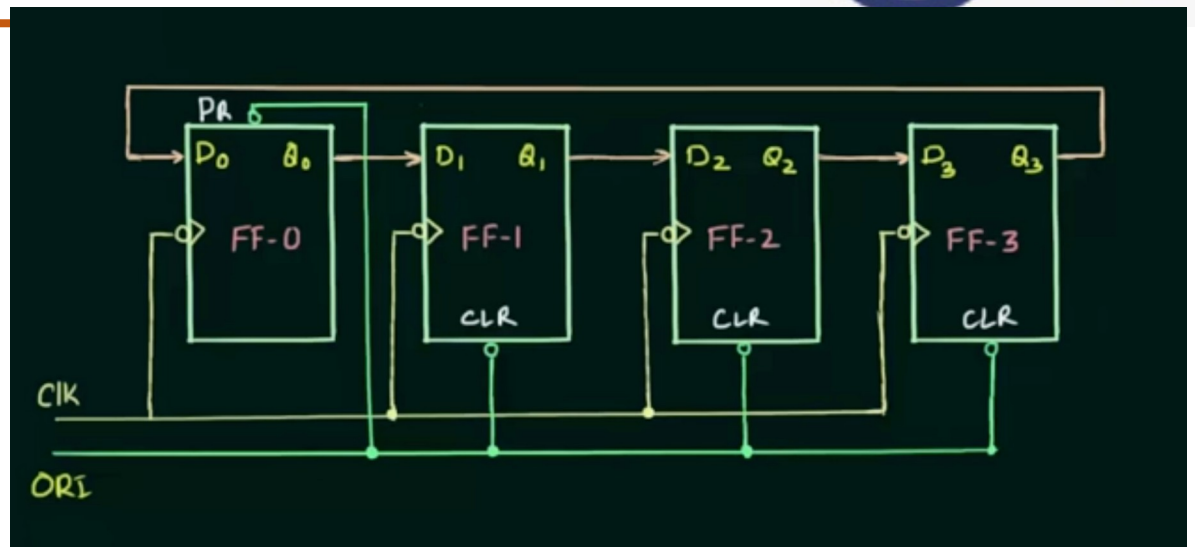
Circuit & Working

4-Bit Ring Counter

- Four flip-flops are connected as a shift register.
- All flip-flops receive a **common clock**.
- The output of the last flip-flop is connected back to the input of the first.
- The counter must be initialized with a single 1.

Working: Q0 → Q1 → Q2 → Q3 → Q0

At every clock pulse, the 1 shifts to the next flip-flop.



ORI is an initialization/reset-type control signal. It forces the ring counter into its starting state, usually 1000, before normal counting begins.

So remember:

ORI = Overriding Input

ORI = 0 → initialize the counter

ORI = 1 → normal ring-counter operation

One important point: **ORI is not the clock.** It is an overriding control used to establish the initial 1 in the ring.

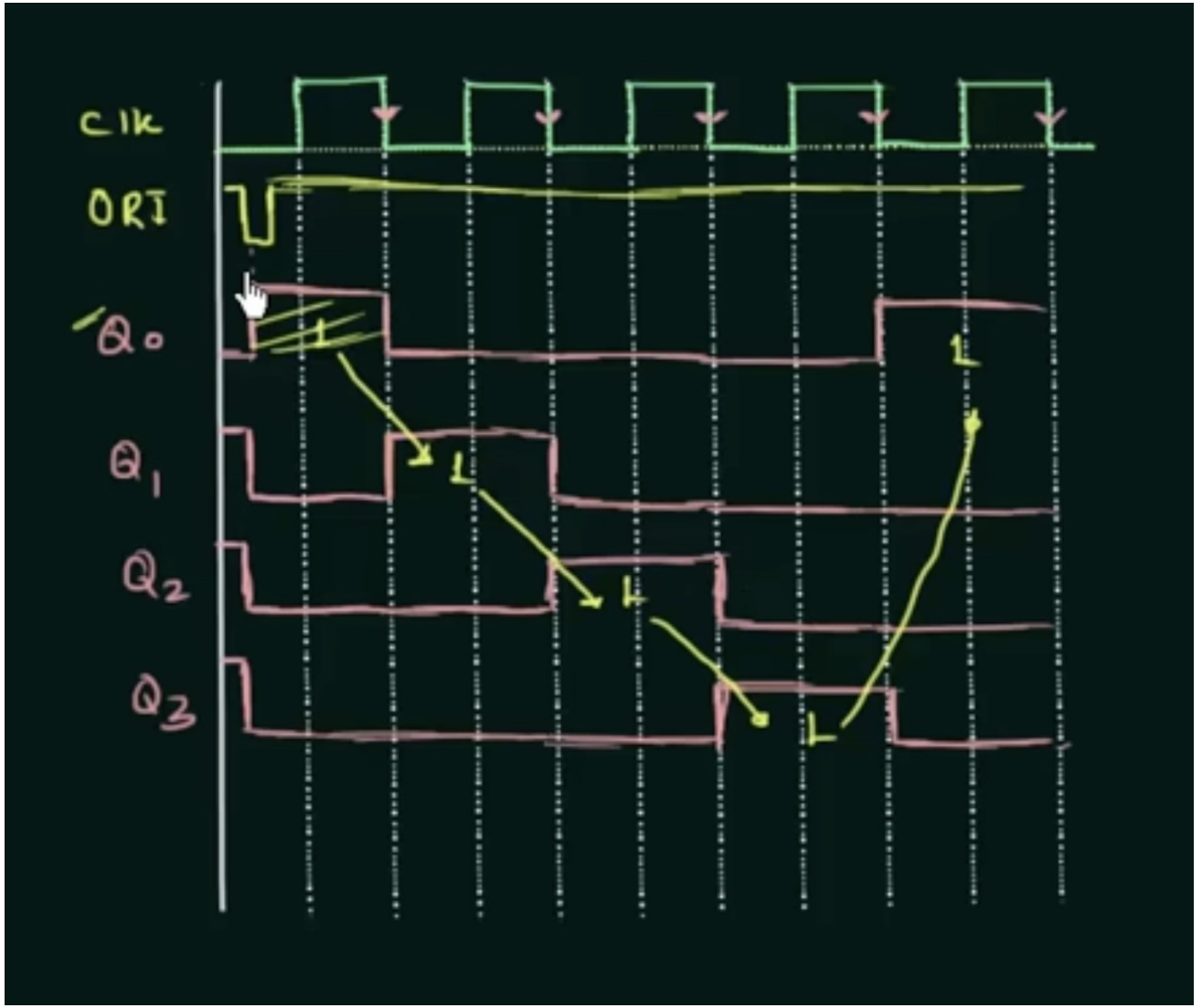
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Type of counters - Ring counters

ORI	CLK	Q_0	Q_1	Q_2	Q_3
⊔	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

Ring Counter Timing/State Table



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Type of counters - Ring counters

Advantages

- Simple circuit design
- Fast operation
- Directly provides decoded outputs
- No additional decoding logic required

Limitations

- Requires **n flip-flops for n states**
- More hardware compared with binary counters
- Requires proper initialization
- 0000 is an invalid/stuck state

Applications

- Sequence generation
- Timing and control circuits
- Stepper motor control
- Digital control systems
- Timing signal generation

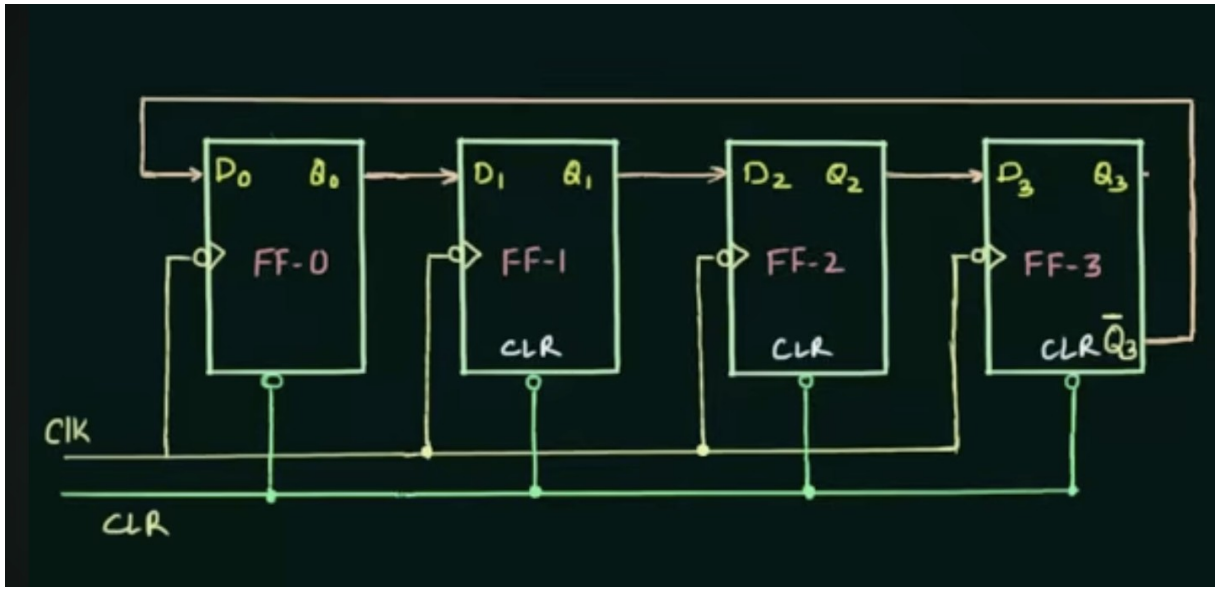
A ring counter circulates a single HIGH bit through a series of flip-flops to generate a repeating sequence of states.

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Type of counters - Johnson counters

- A Johnson counter is a modified **shift-register counter**.
- It is also called a **Twisted Ring Counter** or **Switch-Tail Ring Counter**.
- The **complement of the last flip-flop output** is fed back to the input of the first flip-flop.
- For **n flip-flops, a Johnson counter has 2n valid states**.



Example:

4-bit Johnson Counter → **8 states**

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Type of counters - Johnson counters

4-Bit Johnson Counter

The complemented output of the last flip-flop is fed back to the first.

Feedback:

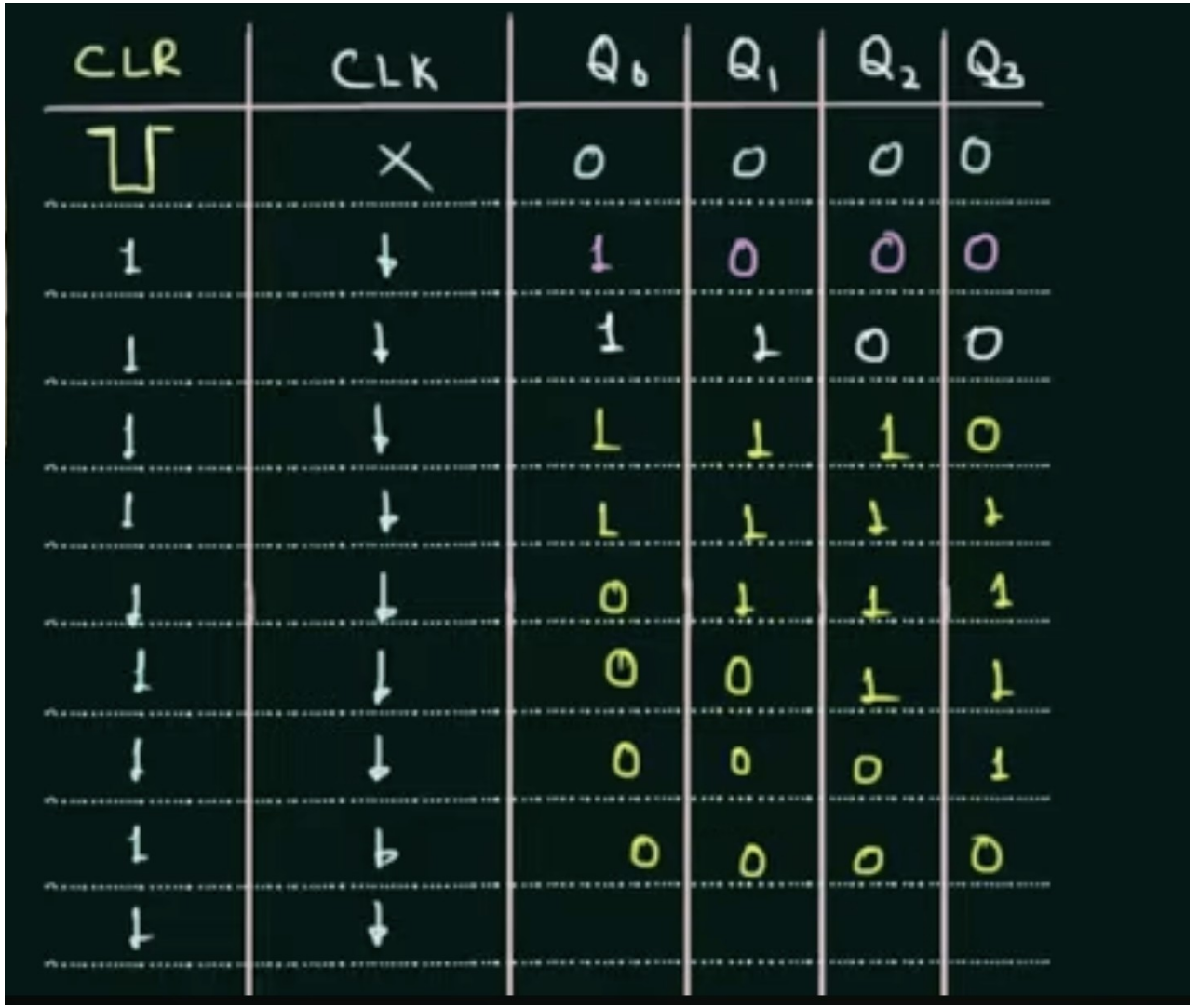
$D_0 = Q_3$^(-)

At every clock pulse, the contents shift by one position.

Starting from 0000:

0000 → 1000 → 1100 → 1110 → 1111 → 0111 → 0011 → 0001 → 0000

Thus, the sequence repeats after **8 clock pulses**.

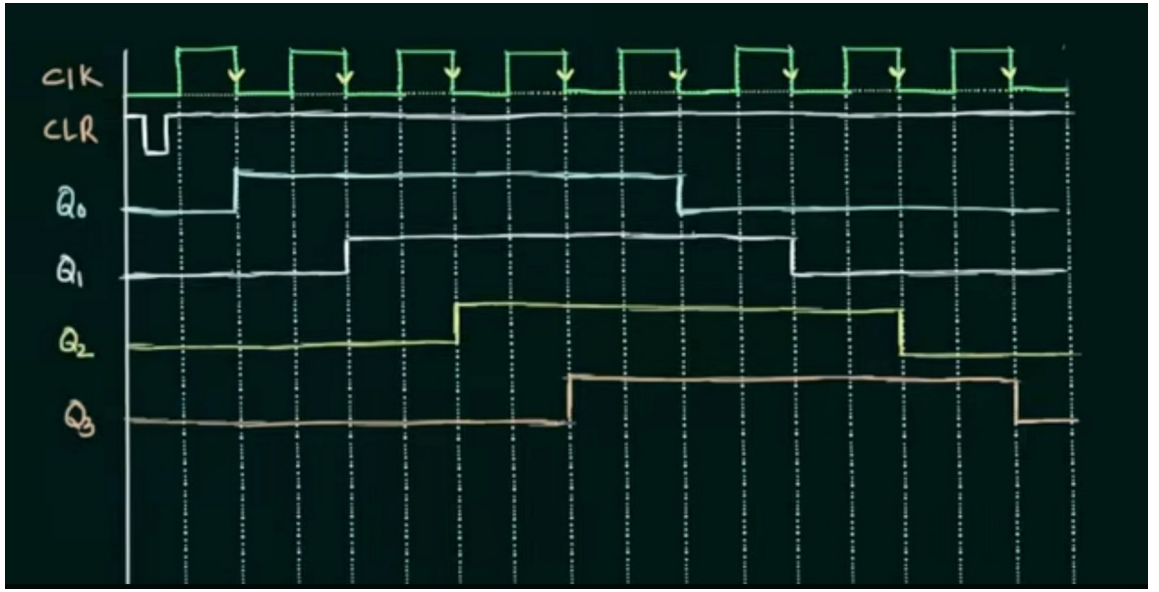


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Type of counters - Johnson counters

CLR	CLK	Q_0	Q_1	Q_2	Q_3
⊓	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	1
1	↓	0	0	0	0
1	↓				



Timing Waveforms of a 4-Bit Johnson (Twisted Ring) Counter

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Worked Example: Testing a 32-bit Counter

- Goal: confirm the counter correctly increments.
- Step 1: Shift in test pattern 0111…111 (32 bits) - 32 cycles.
- Step 2: Switch to normal mode, run 1 clock cycle (counter increments).
- Step 3: Shift the result out - 32 cycles , verify it reads 1000…000.
- **Total: 65 cycles**, instead of ~2 billion.

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Questions

1. An N-bit binary counter with reset advances through how many distinct output values before repeating?

A) N

B) N^2

C) 2N

D) 2^N

2. In the counter design built from an adder and a resettable register, what value is added to the register's contents on every clock cycle?

A) 0

B) 1

C) N

D) Reset value

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Questions

1. An N-bit binary counter with reset advances through how many distinct output values before repeating?

A) N

B) N^2

C) 2N

D) 2^N

Answer: D

2. In the counter design built from an adder and a resettable register, what value is added to the register's contents on every clock cycle?

A) 0

B) 1

C) N

D) Reset value

Answer: B

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Questions

3. In the counter HDL, what has priority when both reset and the clock edge occur?

A) The increment (q <= q + 1)

B) Reset (q <= 0)

C) Neither - output is undefined

D) Load

4. What is the key structural difference between a plain shift register and a shift register with parallel load?

A) The plain version has no clock input

B) The parallel-load version adds a D input and Load control with a mux before each flip-flop

C) The plain version uses combinational logic only

D) The parallel-load version removes Sout

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Questions

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C) The plain version uses combinational logic only

D) The parallel-load version removes Sout

Answer: B

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Questions

5. Why is testing sequential circuits harder than testing combinational circuits?

A) Sequential circuits have no outputs

B) Sequential circuits' outputs depend on internal state reached over many cycles, not just current inputs

C) Sequential circuits cannot be clocked

D) Sequential circuits use more transistors

6. In a scan chain, how does a flip-flop behave in normal (non-test) mode?

A) It loads from its usual D input; scan connections are ignored

B) It always loads from Sin

C) It shifts one position per cycle regardless of mode

D) It resets every cycle

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Questions

5. Why is testing sequential circuits harder than testing combinational circuits?

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C) Sequential circuits cannot be clocked

D) Sequential circuits use more transistors

Answer: B

6. In a scan chain, how does a flip-flop behave in normal (non-test) mode?

A) It loads from its usual D input; scan connections are ignored

B) It always loads from Sin

C) It shifts one position per cycle regardless of mode

D) It resets every cycle

Answer: A

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Questions

7. For the 32-bit counter scan-chain test example, what is the total number of cycles needed (shift in + 1 count cycle + shift out)?

A) 32

B) 33

C) 64

D) 65

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Questions

7. For the 32-bit counter scan-chain test example, what is the total number of cycles needed (shift in + 1 count cycle + shift out)?

A) 32

B) 33

C) 64

D) 65

Answer: D

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THANK YOU

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